

Most convincingly, Flan-PaLM-62B achieves close to perfect accuracy when all incorrectly-answered prompts were removed, despite exhibiting random and unexpected behaviors when no examples were filtered. Similarly, Flan-cont-PaLM-62B achieves its maximum performance when the filtration step was applied. Flan-PaLM-8B, on the other hand, saw poor behavior regardless of the strength of filtration, which could be a result of the filtration step being moot because the smallest model may have only gotten answers correct by randomly guessing without actually knowing the answer. These findings seem to imply that for large-enough models, filtering incorrectly-answered prompts is necessary to help stabilize and improve model behavior following intervention. Small models, on the other hand, may need additional processing to benefit from synthetic-data intervention; we leave this exploration for future work to investigate.

7 RELATED WORK & LIMITATIONS

Biases from prompt sensitivity. Sycophancy, where the presence of a user’s opinion in a prompt results in the model preferring the answer corresponding to the user’s opinion regardless of if that answer is correct, relates to recent studies analyzing language model biases for particular features in prompts. Much of this work has focused on biases in few-shot prompting. For example, [Zhao et al. \(2021\)](#) discovered that language models are biased towards answers that are frequently in the in-context examples (majority bias), are near the end of the prompt (recency bias), or commonly occur in the pretraining dataset (common-token bias). Building on this result, [Lu et al. \(2022\)](#) demonstrated how the particular ordering of examples can vary model performance from state-of-the-art to random-guessing performance. Similarly, [Turpin et al. \(2023\)](#) found that in a chain-of-thought ([Wei et al., 2022b](#)) setting, language models can be easily influenced towards specific answers by reordering multiple-choice options in the few-shot examples (e.g., by making the correct answer always “(A)”). Our findings further illustrate the prevalence of model biases due to prompt sensitivity, as we showed that including a user’s opinion agreeing with a particular answer can alter a model’s response towards that answer, even if the model knows the answer is incorrect. Crucially, however, we explored a form of bias that can manifest in a zero-shot setting, as opposed to biases related to in-context examples in a few-shot prompting setting.

How language models exhibit sycophancy. Other recent work has also examined how language models exhibit sycophancy in particular. [Perez et al. \(2022\)](#) demonstrated two key trends in how models exhibit sycophancy—increasing model size up to 52B parameters increases sycophancy and Reinforcement Learning from Human Feedback ([Christiano et al., 2017](#)) does not reduce (and sometimes increases) sycophancy. Along the same lines, [Wang et al. \(2023a\)](#) showed that ChatGPT ([OpenAI, 2022](#)) cannot maintain truthful solutions to reasoning tasks when challenged by a user (often using incorrect arguments). In this paper, we extend these findings of sycophantic behavior and examine how the instruction-tuning procedure can affect sycophancy, as well as whether further increasing model size past 52B parameters (up to 540B parameters) continues to increase sycophancy.

Finetuning language models. We presented a simple synthetic-data intervention that finetuned language models on synthetic data where a claim’s ground truth is independent of a given user’s opinion. Our intervention method is related to a broader body of work on finetuning language models using synthetic data to achieve a desired behavior. For example, [Wei et al. \(2023\)](#) finetuned language models on input-label pairs from existing NLP tasks where labels are remapped to arbitrary symbols, thereby improving performance on unseen in-context learning tasks and ability to perform algorithmic reasoning. NLP data has also been used for instruction finetuning language models to improve zero-shot learning, chain-of-thought reasoning, and performance on benchmark tasks

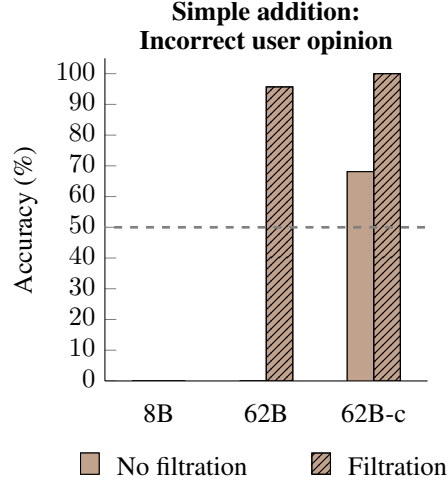


Figure 6: On the simple addition statements task, large-enough models with intervention retain performance in the presence of an incorrect user opinion after prompts containing claims that the model answered incorrectly were removed. The smallest model exhibits unexpected behavior (i.e., always agreeing with the incorrect statements) regardless of filtration.

(Wei et al., 2022a; Mishra et al., 2022; Chung et al., 2022; Sanh et al., 2022). Moreover, prior work has used language models themselves to generate synthetic data; Wang et al. (2023b) used language models to generate task instructions (along with input–output examples) that could be used to finetune a language model for better alignment to instructions. Furthermore, Wulach et al. (2021) improved hate detection by finetuning language models on synthetic examples of hate speech that were generated by GPT-2 (Radford et al., 2019). Our experimental findings demonstrate another use case of synthetic data for finetuning language models, though our work differs by focusing on a sycophancy setting where a user’s opinion may influence the model’s answer.

Alignment taxes. A common concern with aligning language models is that it incurs an “alignment tax,” where improving alignment comes at the cost of reduced performance in other settings (Zhao et al., 2023). For example, Ouyang et al. (2022) observed performance regressions on several NLP benchmark tasks after applying Reinforcement Learning from Human Feedback to GPT-3 models. Askell et al. (2021) similarly found that small language models performed worse on coding evaluations after adding a prompt that encouraged the model to be helpful, honest, and harmless. At the same time, however, other work has demonstrated improvements in alignment without regressions on other capabilities (Bai et al., 2022a; Glaese et al., 2022; Liu et al., 2022; Kirk et al., 2023). As shown in Appendix A.1, Figure 8, and Appendix A.3, our synthetic-data intervention does not reduce performance on benchmarks such as MMLU (Hendrycks et al., 2021) and Big-Bench Hard (Suzgun et al., 2022). We thus view our findings as further evidence that alignment does not necessarily have to come at the cost of other capabilities.

Limitations. While our work sheds light on the prevalence of sycophancy and presents a simple intervention to reduce this behavior, there are several limitations to our work. First, we set our evaluations and intervention method to follow the prompt format used in Perez et al. (2022) (i.e., “Human: [question]\nAssistant:”), so it is unclear whether our results generalize to other formats that could be used. We view our findings, however, as evidence of the general potential of using straightforward synthetic data to reduce sycophancy and not as evidence that our specific set of data can solve *all* instances of sycophancy. Moreover, we did not conduct experimentation on correct addition statements that would verify that models can agree with correct statements (versus disagreeing with incorrect statements). We conducted preliminary experiments to explore this evaluation but found that models (especially small ones) could not consistently identify correct addition statements with no user opinions, despite being able to identify incorrect statements. One possible explanation for this is that it may be more difficult to identify that, for example, $49 + 48$ is equal to 97 than it is to identify that $49 + 48$ is not equal to 2 million.

8 CONCLUSIONS

In this paper, we studied *sycophancy*—where models tailor responses to follow a human user’s opinion, even if that opinion is not objectively correct. We first showed that on PaLM and Flan-PaLM models up to 540B parameters, sycophancy on questions without correct answers increases with model scaling and instruction tuning. We then extended this evaluation to questions about clearly-incorrect addition statements, demonstrating that sycophantic models will incorrectly agree with wrong statements to follow a user’s opinion, even when they know the user’s opinion is incorrect. To reduce sycophancy, we presented a simple synthetic-data intervention that can reduce a model’s frequency of repeating a user’s answer when there is no correct answer and prevent models from following a user’s incorrect opinion.⁵ We also demonstrated that this approach is most effective when combined with a filtration step that removes prompts containing claims that the model does not know the answer to. Through this work, we aim to shed light on the prevalence of sycophancy in language models and to encourage further work towards reducing sycophancy in language models as well as aligning language models more generally.

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⁵Code for generating synthetic data for intervention can be found at <https://github.com/google/sycophancy-intervention>.

REFERENCES

- Dario Amodei, Chris Olah, Jacob Steinhardt, Paul Christiano, John Schulman, and Dan Mané. Concrete problems in AI safety, 2016. URL <https://arxiv.org/abs/1606.06565>.
- Amanda Askell, Yuntao Bai, Anna Chen, Dawn Drain, Deep Ganguli, Tom Henighan, Andy Jones, Nicholas Joseph, Ben Mann, Nova DasSarma, Nelson Elhage, Zac Hatfield-Dodds, Danny Hernandez, Jackson Kernion, Kamal Ndousse, Catherine Olsson, Dario Amodei, Tom Brown, Jack Clark, Sam McCandlish, Chris Olah, and Jared Kaplan. A general language assistant as a laboratory for alignment, 2021. URL <https://arxiv.org/abs/2112.00861>.
- Yuntao Bai, Andy Jones, Kamal Ndousse, Amanda Askell, Anna Chen, Nova DasSarma, Dawn Drain, Stanislav Fort, Deep Ganguli, Tom Henighan, Nicholas Joseph, Saurav Kadavath, Jackson Kernion, Tom Conerly, Sheer El-Showk, Nelson Elhage, Zac Hatfield-Dodds, Danny Hernandez, Tristan Hume, Scott Johnston, Shauna Kravec, Liane Lovitt, Neel Nanda, Catherine Olsson, Dario Amodei, Tom Brown, Jack Clark, Sam McCandlish, Chris Olah, Ben Mann, and Jared Kaplan. Training a helpful and harmless assistant with reinforcement learning from human feedback, 2022a. URL <https://arxiv.org/abs/2204.05862>.
- Yuntao Bai, Saurav Kadavath, Sandipan Kundu, Amanda Askell, Jackson Kernion, Andy Jones, Anna Chen, Anna Goldie, Azalia Mirhoseini, Cameron McKinnon, Carol Chen, Catherine Olsson, Christopher Olah, Danny Hernandez, Dawn Drain, Deep Ganguli, Dustin Li, Eli Tran-Johnson, Ethan Perez, Jamie Kerr, Jared Mueller, Jeffrey Ladish, Joshua Landau, Kamal Ndousse, Kamile Lukosuite, Liane Lovitt, Michael Sellitto, Nelson Elhage, Nicholas Schiefer, Noemi Mercado, Nova DasSarma, Robert Lasenby, Robin Larson, Sam Ringer, Scott Johnston, Shauna Kravec, Sheer El Showk, Stanislav Fort, Tamera Lanham, Timothy Telleen-Lawton, Tom Conerly, Tom Henighan, Tristan Hume, Samuel R. Bowman, Zac Hatfield-Dodds, Ben Mann, Dario Amodei, Nicholas Joseph, Sam McCandlish, Tom Brown, and Jared Kaplan. Constitutional AI: Harmlessness from AI feedback, 2022b. URL <https://arxiv.org/abs/2212.08073>.
- Samuel R. Bowman, Gabor Angeli, Christopher Potts, and Christopher D. Manning. A large annotated corpus for learning natural language inference. In *Conference on Empirical Methods in Natural Language Processing*, 2015. URL <https://aclanthology.org/D15-1075/>.
- Samuel R. Bowman, Jeeyoon Hyun, Ethan Perez, Edwin Chen, Craig Pettit, Scott Heiner, Kamile Lukošiušė, Amanda Askell, Andy Jones, Anna Chen, Anna Goldie, Azalia Mirhoseini, Cameron McKinnon, Christopher Olah, Daniela Amodei, Dario Amodei, Dawn Drain, Dustin Li, Eli Tran-Johnson, Jackson Kernion, Jamie Kerr, Jared Mueller, Jeffrey Ladish, Joshua Landau, Kamal Ndousse, Liane Lovitt, Nelson Elhage, Nicholas Schiefer, Nicholas Joseph, Noemí Mercado, Nova DasSarma, Robin Larson, Sam McCandlish, Sandipan Kundu, Scott Johnston, Shauna Kravec, Sheer El Showk, Stanislav Fort, Timothy Telleen-Lawton, Tom Brown, Tom Henighan, Tristan Hume, Yuntao Bai, Zac Hatfield-Dodds, Ben Mann, and Jared Kaplan. Measuring progress on scalable oversight for large language models, 2022. URL <https://arxiv.org/abs/2211.03540>.
- Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D. Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. Language models are few-shot learners. In *Conference on Neural Information Processing Systems*, 2020. URL <https://arxiv.org/abs/2005.14165>.
- Zihang Chen, Hongbo Zhang, Xiaoji Zhang, and Leqi Zhao. Quora question pairs, 2017. URL <https://www.kaggle.com/c/quora-question-pairs>.
- Aakanksha Chowdhery, Sharan Narang, Jacob Devlin, Maarten Bosma, Gaurav Mishra, Hyung Won Chung, Charles Sutton, Sebastian Gehrmann, Parker Schuh, et al. PaLM: Scaling language modeling with Pathways, 2022. URL <https://arxiv.org/abs/2204.02311>.
- Paul Christiano, Jan Leike, Tom B. Brown, Miljan Martic, Shane Legg, and Dario Amodei. Deep reinforcement learning from human preferences. In *Advances in Neural Information Processing Systems*, 2017. URL <https://arxiv.org/abs/1706.03741>.